

SUPPLY MANAGEMENT SYSTEM — Functional Specification Document

FSD Addendum 33 — Fulfillment Routes & Sale Order Delivery Integration

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Architecture	.NET 8 / EF Core 8 / SQL Server 2022 / React 18
Change Items	5 changes (C1-C5)
Impact	4 migrations, 2 modified entities, 3 new tables, 5 new/modified UI panels, 1 Hangfire job

Table of Contents

- 1. [Purpose & Scope](#)
- 2. [Existing Delivery Module Confirmation](#)
- 3. [C1 — Fulfillment Route Configuration](#)
- 4. [C2 — Product Variant Route Assignment](#)
- 5. [C3 — Sale Order Line Route Selection & Confirmation Gate](#)
- 6. [C4 — Delivery Order Creation from Sale Order](#)
- 7. [C5 — Route-Aware State Machine Transitions](#)
- 8. [Database Migrations](#)
- 9. [API Changes](#)

- 10. UI Wireframes & Specifications
 - 11. Business Rules
 - 12. Test Scenarios
 - 13. Development Phases
-

1. Purpose & Scope

This addendum introduces Fulfillment Routes — a configurable mechanism that determines the warehouse operations required to fulfill each product on a Sale Order.

Fulfillment Routes are inspired by Odoo's route system and define whether a product requires picking only (customer collects), pick-and-ship, or the full pick-pack-ship cycle.

1.1 Design Principles

- **Route-Driven Fulfillment** — The fulfillment path (which warehouse steps are executed) is determined by the route, not hard-coded. Routes are defined per organization and assigned at the product variant level.
- **Override at Sale Order** — Each SO line can override the variant's default route, giving salespeople flexibility when the customer's delivery needs differ from the norm.
- **Confirmation Gate** — A Sale Order cannot be confirmed unless every line has a fulfillment route, either inherited from the product variant or explicitly set on the SO line. This ensures the warehouse always knows what to do.
- **Reuse Existing Infrastructure** — Delivery orders, pick lists, shipment packages, and the 15-status state machine already exist. This addendum extends them, not replaces them.

1.2 Changes Summary

Change ID	Title	Category
C1	Fulfillment Route Configuration	New entity — FulfillmentRoutes & FulfillmentRouteSteps
C2	Product Variant Route Assignment	Modify entity — add fulfillment_route_id to ProductVariants
C3	Sale Order Line Route Selection & Confirmation Gate	Modify entity — add fulfillment_route_id to SaleOrderLines + validation
C4	Delivery Order Creation from Sale Order	New service — SO confirmation triggers delivery order(s) grouped by route
C5	Route-Aware State Machine Transitions	Modify service — delivery orders skip steps not in their route

2. Existing Delivery Module Confirmation

The following logistics module entities and services are already implemented and are the foundation for this addendum. No changes are needed to these structures — only extensions.

2.1 Existing Schema (logistics)

Entity	Purpose	Key Fields
delivery_orders	Header for outbound/inbound delivery	delivery_order_id, org_id, from_source_type (PO/SRO/MIV/Transfer), status (15-state machine), warehouse_id
delivery_order_lines	Line items linking to source documents	delivery_order_line_id, delivery_order_id, variant_id, quantity, fulfilled_qty
pick_lists	Generated picking instructions	pick_list_id, delivery_order_id, FEFO allocation
pick_list_lines	Per-line pick instructions with bin locations	pick_list_line_id, zone, bin, lot_id, quantity
shipment_packages	Handling units (cartons, pallets)	package_id, hu_barcode, parent_package_id (nesting)
package_contents	Links packages to delivery lines	package_content_id, package_id, delivery_order_line_id, quantity
consignments	Carrier booking unit	consignment_id, carrier_account_id, 16-status state machine
consignment_stops	Pickup/delivery waypoints	stop_id, consignment_id, stop_type, address_id

2.2 Existing Delivery State Machine (15 Statuses)

DRAFT → RELEASED → PICKING → PICKED → PACKED → STAGED
→ PENDING_APPROVAL → GOODS_ISSUED → IN_TRANSIT → AT_HUB
→ OUT_FOR_DELIVERY → DELIVERY_ATTEMPTED → DELIVERED
→ COMPLETED → CANCELLED

2.3 Existing from_source_type Values

Source Type	Description	Currently Implemented
PO	Purchase Order receipt	✓ Yes
SRO	Stock Return Order	✓ Yes
MIV	Manual Issue Voucher	✓ Yes
Transfer	Inter-warehouse transfer	✓ Yes
SALE_ORDER	Sale Order outbound delivery	✗ To be added in this addendum

2.4 Existing IStockReservationService

```
csharp

// Already implemented in SMS.Shared – no changes needed
public interface IStockReservationService
{
    Task<ReservationResult> Reserve(ReserveRequest request);
    Task Release(long reservationId);
    Task Consume(long reservationId);
}

// ReservationSourceType enum already includes:
// MIR, DELIVERY, SALES_ORDER
```

2.5 Existing Permissions (Logistics)

Permission	Description
DELIVERY_VIEW	View delivery orders
DELIVERY_EDIT	Create/edit delivery orders
DELIVERY_RELEASE	Release delivery for picking
DELIVERY_CANCEL	Cancel a delivery order
PICKING	Access pick list
PICK_CONFIRM	Confirm pick quantities
PACKING	Access packing station
PACK_CONFIRM	Confirm packed items
DISPATCH	Dispatch for shipment
GOODS_ISSUE	Post goods issue (accounting event)

 **NOTE:** All existing logistics entities, state machines, services, and permissions are confirmed as-is. This addendum adds to them without modifying existing behavior.

3. C1 — Fulfillment Route Configuration

3.1 Overview

A Fulfillment Route is an organization-defined template that specifies which warehouse operations are required to fulfill a product. Each route has an ordered list of steps drawn from the existing delivery state machine.

Three seed routes cover the most common patterns:

Route	Steps	Use Case
PICK_ONLY	Pick	Customer collects from warehouse; no packing or shipping
PICK_AND_SHIP	Pick → Goods Issue → Ship	Standard items that don't need packing
PICK_PACK_SHIP	Pick → Pack → Goods Issue → Ship	Items requiring packing (boxing, palletizing) before shipment

Organizations can create custom routes beyond these three seeds.

3.2 Schema — FulfillmentRoutes

New Table: `logistics.FulfillmentRoutes`

Column	Type	Nullable	Default	Description
fulfillment_route_id	INT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants.Organizations	Tenant discriminator
code	NVARCHAR(30)	NOT NULL		Short code (e.g., PICK_ONLY, PICK_AND_SHIP)
name	NVARCHAR(100)	NOT NULL		Display name
description	NVARCHAR(500)	NULL		Explanation of when to use this route
is_default	BIT	NOT NULL	0	Whether this is the org's default route (at most one per org)
is_active	BIT	NOT NULL	1	Soft delete
is_system	BIT	NOT NULL	0	TRUE for seed routes — cannot be deleted
requires_packing	BIT	NOT NULL	0	Computed convenience flag — TRUE if route includes PACK step
requires_shipping	BIT	NOT NULL	0	Computed convenience flag — TRUE if route includes SHIP step
display_order	INT	NOT NULL	0	Sort order in dropdowns
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Column	Type	Nullable	Default	Description
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Indexes:

Index	Columns	Type
UQ_FulfillmentRoutes_OrgCode	org_id, code	Unique
IX_FulfillmentRoutes_OrgDefault	org_id, is_default	Filtered (WHERE is_default = 1)

3.3 Schema — FulfillmentRouteSteps

New Table: `logistics.FulfillmentRouteSteps`

Column	Type	Nullable	Default	Description
route_step_id	INT IDENTITY	NOT NULL	PK	Primary key
fulfillment_route_id	INT	NOT NULL	FK → FulfillmentRoutes	Parent route
org_id	INT	NOT NULL	FK → tenants.Organizations	Tenant discriminator
step_code	NVARCHAR(20)	NOT NULL		One of: PICK, PACK, STAGE, APPROVAL, GOODS_ISSUE, SHIP
step_order	INT	NOT NULL		Execution order (1, 2, 3...)
is_mandatory	BIT	NOT NULL	1	Whether this step can be skipped at runtime
description	NVARCHAR(200)	NULL		Step description or instructions

Indexes:

Index	Columns	Type
UQ_FulfillmentRouteSteps_RouteOrder	fulfillment_route_id, step_order	Unique
UQ_FulfillmentRouteSteps_RouteStep	fulfillment_route_id, step_code	Unique

3.4 Step Code Definitions

Each step_code maps to a segment of the existing 15-status delivery state machine:

step_code	Delivery Statuses Involved	Description
PICK	RELEASED → PICKING → PICKED	Generate pick list, warehouse worker picks items
PACK	PICKED → PACKED	Pack picked items into handling units (shipment_packages)
STAGE	PACKED → STAGED	Move packed items to staging/loading area
APPROVAL	STAGED → PENDING_APPROVAL	Manager approval before dispatch (optional)
GOODS_ISSUE	(previous) → GOODS_ISSUED	Post goods issue — stock decremented, COGS posted
SHIP	GOODS_ISSUED → IN_TRANSIT → ... → DELIVERED	Create consignment, carrier booking, tracking

3.5 Seed Route Definitions

PICK_ONLY — Customer Collects

step_order	step_code	is_mandatory
1	PICK	YES
2	GOODS_ISSUE	YES

State machine path: DRAFT → RELEASED → PICKING → PICKED → GOODS_ISSUED → COMPLETED

Skipped statuses: PACKED, STAGED, PENDING_APPROVAL, IN_TRANSIT through DELIVERED (customer takes physical possession at warehouse).

PICK_AND_SHIP — Standard Shipment (No Packing)

step_order	step_code	is_mandatory
1	PICK	YES
2	GOODS_ISSUE	YES
3	SHIP	YES

State machine path: DRAFT → RELEASED → PICKING → PICKED → GOODS_ISSUED → IN_TRANSIT → ...
→ DELIVERED → COMPLETED

Skipped statuses: PACKED, STAGED, PENDING_APPROVAL.

PICK_PACK_SHIP — Full Cycle

step_order	step_code	is_mandatory
1	PICK	YES
2	PACK	YES
3	GOODS_ISSUE	YES
4	SHIP	YES

State machine path: DRAFT → RELEASED → PICKING → PICKED → PACKED → GOODS_ISSUED →
IN_TRANSIT → ... → DELIVERED → COMPLETED

Skipped statuses: STAGED, PENDING_APPROVAL.

 **NOTE:** Organizations can create custom routes with additional steps. For example, a route with STAGE and APPROVAL for high-value items: PICK → PACK → STAGE → APPROVAL → GOODS_ISSUE → SHIP (uses all 15 statuses).

4. C2 — Product Variant Route Assignment

4.1 Overview

Each product variant can have a default fulfillment route. This tells the system "when this product is sold, here's how it should be fulfilled." The route assignment lives on the product variant because different variants of the same product may require different handling (e.g., a 1kg pack ships directly, but a 500kg bulk variant requires packing onto pallets).

4.2 Schema Change — ProductVariants

Modify: `inventory.ProductVariants`

ProductVariants — New Column (ADD)

Column	Type	Nullable	Default	Description
fulfillment_route_id	INT	NULL	FK → logistics.FulfillmentRoutes	Default fulfillment route for this variant

New Index:

Index	Columns	Type
IX_ProductVariants_FulfillmentRoute	fulfillment_route_id	Non-unique, filtered (WHERE NOT NULL)

4.3 UI — Product Variant Form

The fulfillment route is set on the product variant form in the existing "Logistics" or "Inventory" tab:

Product Variant: Steel Rod 10mm - 6M Length

[General] [Pricing] [Inventory] [Logistics] [History]

▲ active

Fulfillment Route

Pick & Ship (PICK_AND_SHIP) ▼

Steps: Pick → Goods Issue → Ship

① This route determines the warehouse operations when this variant is sold. Can be overridden per Sale Order line.

Weight & Dimensions

Net Wt

12.5 kg

Gross Wt

13.0 kg

L

6 m

W

10mm

H

10mm

Storage

Default Zone

Zone A – Racks

Storage Class

Standard

4.4 Bulk Assignment

For organizations setting up routes across many variants, a bulk assignment endpoint allows setting the route by product category:

```
POST /api/fulfillment-routes/{routeId}/assign-by-category
```

```
Body: { "product_category_id": 15 }
```

This sets fulfillment_route_id on all variants under that category that currently have NULL (does not overwrite existing assignments).

5. C3 — Sale Order Line Route Selection & Confirmation Gate

5.1 Overview

Two changes to Sale Order lines:

- 1. **Route Column** — Each SO line shows the effective fulfillment route (inherited from variant or overridden) and allows the salesperson to change it.
- 2. **Confirmation Gate** — The SO cannot transition from DRAFT to CONFIRMED unless every line has a resolved fulfillment route.

5.2 Schema Change — SaleOrderLines

Modify: `demand.SaleOrderLines`

SaleOrderLines — New Column (ADD)

Column	Type	Nullable	Default	Description
fulfillment_route_id	INT	NULL	FK → logistics.FulfillmentRoutes	Route override for this line. NULL = inherit from variant.

New Index:

Index	Columns	Type
IX_SaleOrderLines_FulfillmentRoute	fulfillment_route_id	Non-unique, filtered (WHERE NOT NULL)

5.3 Effective Route Resolution

The effective route for a SO line is resolved in this priority order:

`csharp`

```
// Pseudocode – EffectiveRouteResolver
public int? ResolveEffectiveRouteId(SaleOrderLine line)
{
    // Priority 1: Explicit override on the SO line
    if (line.FulfillmentRouteId.HasValue)
        return line.FulfillmentRouteId;

    // Priority 2: Default route on the product variant
    var variant = _variantRepo.GetById(line.VariantId);
    if (variant.FulfillmentRouteId.HasValue)
        return variant.FulfillmentRouteId;

    // Priority 3: Organization default route
    var orgDefault = _routeRepo.GetOrgDefault(line.OrgId);
    if (orgDefault != null)
        return orgDefault.FulfillmentRouteId;

    // No route found – this line will block confirmation
    return null;
}
```

Resolution priority:

Priority	Source	Description
1 (highest)	SO Line override	Salesperson explicitly chose a route for this line
2	Product Variant default	Route configured on the variant master data
3	Organization default route	The route with is_default = 1 for this org
4	NULL — blocks confirmation	No route resolved anywhere

5.4 Confirmation Gate Validation

When a user attempts to confirm a Sale Order (DRAFT → CONFIRMED), the system validates:

csharp

```

// Pseudocode - SaleOrderConfirmationValidator
public ValidationResult ValidateForConfirmation(SaleOrder order)
{
    var errors = new List<ValidationError>();
    var missingRouteLines = new List<int>();

    foreach (var line in order.Lines)
    {
        var effectiveRouteId = _routeResolver.ResolveEffectiveRouteId(line);

        if (effectiveRouteId == null)
        {
            missingRouteLines.Add(line.LineNumber);
        }
        else
        {
            // Validate the route is still active
            var route = _routeRepo.GetById(effectiveRouteId.Value);
            if (route == null || !route.IsActive)
            {
                errors.Add(new ValidationError(
                    $"Line {line.LineNumber}: fulfillment route '{route?.Code}' is
                ));
            }
        }
    }

    if (missingRouteLines.Any())
    {
        errors.Add(new ValidationError(
            $"Cannot confirm: lines {string.Join(", ", missingRouteLines)} " +
            $"have no fulfillment route. Assign a route on each line or " +
            $"set a default route on the product variant."));
    }

    return new ValidationResult(errors);
}

```

Confirmation is blocked when:

- Any SO line has no effective route (not on line, not on variant, no org default)
- Any SO line's effective route is inactive or deleted

Confirmation proceeds when:

- Every SO line resolves to an active fulfillment route (from any of the three priority levels)

5.5 UI — Sale Order Lines with Route Column

Sale Order: SO-1085Status: DRAFT

[Header] [Lines] [Fulfillment] [Attachments] [History]

▲ active

	Product	Qty	Price	Route	Action
1	Steel Pipes	500	\$450	[Pick Only	▼] ⑤
	↳ Source: variant default				
2	Copper Wire	600	\$820	[Pick Pack Ship	▼] 🔗
	↳ Source: overridden on SO line				
3	Synth Hyd Oil	100	\$2800	[Pick & Ship	▼] ⑤
	↳ Source: variant default				
4	Custom Gasket	50	\$150	[– Select Route –	▼] ▲
	↳ ▲ No route defined – order cannot be confirmed				

Legend: ⑤ = inherited from variant 🔗 = overridden on line
▲ = missing route (blocking)

[Confirm Order] ⬅️ DISABLED: 1 line missing fulfillment route

Delivery Preview:

On confirmation, 3 delivery orders will be created:

- DO-1: Steel Pipes (Pick Only – customer collects)
- DO-2: Copper Wire (Pick → Pack → Goods Issue → Ship)
- DO-3: Synth Hyd Oil (Pick → Goods Issue → Ship)
- Line 4: ▲ unroutable – assign route first

⚠ CRITICAL: The "Confirm Order" button is DISABLED and shows a tooltip explaining which lines are missing routes. It is not hidden — the salesperson needs to see it and understand what's blocking them.

5.6 Delivery Preview

Before confirmation, the UI shows a "Delivery Preview" panel that groups SO lines by their effective route and shows how many delivery orders will be created. This preview is computed on the fly (not persisted) and updates as the salesperson changes routes on lines.

6. C4 — Delivery Order Creation from Sale Order

6.1 Overview

When a Sale Order is confirmed, the system automatically creates Delivery Orders grouped by fulfillment route. Lines with the same route go into the same Delivery Order. This extends the existing `delivery_orders` entity with `from_source_type = 'SALE_ORDER'`.

6.2 Schema Change — DeliveryOrders

Modify: `logistics.delivery_orders`

`delivery_orders` — Extend `from_source_type` ENUM

The existing `from_source_type` column (NVARCHAR) accepts a new value:

Value	Description	Direction
PO	Purchase Order receipt (existing)	Inbound
SRO	Stock Return Order (existing)	Inbound
MIV	Manual Issue Voucher (existing)	Outbound
Transfer	Inter-warehouse transfer (existing)	Both
SALE_ORDER	Sale Order outbound delivery (new)	Outbound

`delivery_orders` — New Columns (ADD)

Column	Type	Nullable	Default	Description
sale_order_id	BIGINT	NULL	FK → demand.SaleOrders	Originating Sale Order (NULL for non-SO deliveries)
fulfillment_route_id	INT	NULL	FK → logistics.FulfillmentRoutes	Route governing this delivery's steps

New Indexes:

Index	Columns	Type
IX_DeliveryOrders_SaleOrder	sale_order_id	Non-unique, filtered (WHERE sale_order_id IS NOT NULL)
IX_DeliveryOrders_Route	fulfillment_route_id	Non-unique, filtered (WHERE NOT NULL)

delivery_order_lines — New Column (ADD)

Column	Type	Nullable	Default	Description
sale_order_line_id	BIGINT	NULL	FK → demand.SaleOrderLines	Originating SO line (NULL for non-SO deliveries)

6.3 Delivery Order Creation Logic

csharp

```

// Pseudocode - SaleOrderDeliveryCreator
public async Task<List<DeliveryOrder>> CreateDeliveriesFromSaleOrder(SaleOrder order)
{
    // Group SO lines by their effective fulfillment route
    var linesByRoute = order.Lines
        .GroupBy(l => _routeResolver.ResolveEffectiveRouteId(l)!.Value)
        .ToList();

    var deliveries = new List<DeliveryOrder>();

    foreach (var group in linesByRoute)
    {
        var route = await _routeRepo.GetByIdAsync(group.Key);

        var delivery = new DeliveryOrder
        {
            OrgId = order.OrgId,
            FromSourceType = "SALE_ORDER",
            SaleOrderId = order.SaleOrderId,
            FulfillmentRouteId = route.FulfillmentRouteId,
            WarehouseId = order.WarehouseId,
            Status = "DRAFT",
            ExpectedDate = group.Min(l => l.RequestedDeliveryDate),
            PartnerId = order.PartnerId,
            DeliveryAddressId = order.DeliveryAddressId
        };

        foreach (var soLine in group)
        {
            delivery.Lines.Add(new DeliveryOrderLine
            {
                VariantId = soLine.VariantId,
                Quantity = soLine.Quantity,
                SaleOrderLineId = soLine.SaleOrderLineId,
                OrgId = order.OrgId
            });
        }

        deliveries.Add(delivery);
    }
}

```

```
    await _deliveryRepo.CreateBatchAsync(deliveries);

    // Update SO with delivery reference
    order.HasDeliveries = true;

    return deliveries;
}
```

Grouping Rules:

Scenario	Lines	Delivery Orders Created
All lines same route	3 lines, all PICK_AND_SHIP	1 delivery order with 3 lines
Mixed routes	1× PICK_ONLY, 1× PICK_PACK_SHIP, 1× PICK_AND_SHIP	3 delivery orders, 1 line each
Two lines same route, one different	2× PICK_AND_SHIP, 1× PICK_PACK_SHIP	2 delivery orders (2 lines + 1 line)

6.4 Punjab Group Example

Sale Order SO-1085 confirmed

- └─ Line 1: Steel Pipes 500 @ \$450 Route: PICK_ONLY
- └─ Line 2: Copper Wire 600 @ \$820 Route: PICK_PACK_SHIP
- └─ Line 3: Synth Hyd Oil 100 @ \$2800 Route: PICK_AND_SHIP

System creates 3 Delivery Orders:

DO-20261003-001 (Route: PICK_ONLY)

- └─ from_source_type = SALE_ORDER
- └─ sale_order_id = SO-1085
- └─ fulfillment_route_id = PICK_ONLY
- └─ Line: Steel Pipes × 500
- └─ Steps: Pick → Goods Issue → Complete
(customer sends truck to collect)

DO-20261003-002 (Route: PICK_PACK_SHIP)

- └─ from_source_type = SALE_ORDER
- └─ sale_order_id = SO-1085
- └─ fulfillment_route_id = PICK_PACK_SHIP
- └─ Line: Copper Wire × 600
- └─ Steps: Pick → Pack (spool into boxes) → Goods Issue → Ship → Deliver
(carrier booked, tracking generated)

DO-20261003-003 (Route: PICK_AND_SHIP)

- └─ from_source_type = SALE_ORDER
- └─ sale_order_id = SO-1085
- └─ fulfillment_route_id = PICK_AND_SHIP
- └─ Line: Synth Hyd Oil × 100
- └─ Steps: Pick (drums from yard) → Goods Issue → Ship → Deliver
(no packing needed – drums ship as-is)

6.5 Sale Order ↔ Delivery Order Linkage

```
demand.SaleOrders — 1:N —> logistics.delivery_orders
demand.SaleOrderLines — 1:1 —> logistics.delivery_order_lines
```

The SO detail page's "Fulfillment" tab shows all delivery orders with their statuses. Each delivery order links back to its SO.

7. C5 — Route-Aware State Machine Transitions

7.1 Overview

The existing 15-status delivery state machine remains unchanged. What changes is which statuses a delivery order *passes through* based on its fulfillment route. Statuses not in the route are skipped — the state machine jumps over them.

7.2 Step-to-Status Mapping

Each step_code in the route maps to the delivery statuses it covers. When a step is not in the route, its statuses are skipped:

step_code	Statuses Covered	If Skipped
PICK	RELEASED, PICKING, PICKED	✗ Never skipped — always required
PACK	PACKED	Jump from PICKED directly to next step
STAGE	STAGED	Jump from PACKED (or PICKED) to next step
APPROVAL	PENDING_APPROVAL	Jump from STAGED (or previous) to next step
GOODS_ISSUE	GOODS_ISSUED	✗ Never skipped — always required
SHIP	IN_TRANSIT, AT_HUB, OUT_FOR_DELIVERY, DELIVERY_ATTEMPTED, DELIVERED	Jump from GOODS_ISSUED directly to COMPLETED

7.3 Transition Logic

csharp

```

// Pseudocode - RouteAwareDeliveryStateMachine
public string GetNextStatus(DeliveryOrder delivery, string currentStatus)
{
    var route = _routeRepo.GetWithSteps(delivery.FulfillmentRouteId);
    var routeStepCodes = route.Steps
        .OrderBy(s => s.StepOrder)
        .Select(s => s.StepCode)
        .ToHashSet();

    // Standard transition map (full 15-status chain)
    var fullChain = new[]
    {
        "DRAFT", "RELEASED", "PICKING", "PICKED",
        "PACKED", "STAGED", "PENDING_APPROVAL",
        "GOODS_ISSUED", "IN_TRANSIT", "AT_HUB",
        "OUT_FOR_DELIVERY", "DELIVERY_ATTEMPTED",
        "DELIVERED", "COMPLETED"
    };

    // Map statuses to their step_code
    var statusToStep = new Dictionary<string, string>
    {
        ["RELEASED"] = "PICK", ["PICKING"] = "PICK", ["PICKED"] = "PICK",
        ["PACKED"] = "PACK",
        ["STAGED"] = "STAGE",
        ["PENDING_APPROVAL"] = "APPROVAL",
        ["GOODS_ISSUED"] = "GOODS_ISSUE",
        ["IN_TRANSIT"] = "SHIP", ["AT_HUB"] = "SHIP",
        ["OUT_FOR_DELIVERY"] = "SHIP", ["DELIVERY_ATTEMPTED"] = "SHIP",
        ["DELIVERED"] = "SHIP"
    };

    // Find current position in full chain
    var currentIndex = Array.IndexOf(fullChain, currentStatus);

    // Walk forward, skipping statuses whose step is not in the route
    for (int i = currentIndex + 1; i < fullChain.Length; i++)
    {
        var candidateStatus = fullChain[i];

        // DRAFT and COMPLETED are always valid
    }
}

```

```

    if (candidateStatus == "COMPLETED")
        return candidateStatus;

    // Check if this status's step is in the route
    if (statusToStep.TryGetValue(candidateStatus, out var stepCode))
    {
        if (routeStepCodes.Contains(stepCode))
            return candidateStatus;
        // else: skip this status, continue to next
    }
}

return currentStatus; // should not reach here
}

```

7.4 Route Path Examples

PICK_ONLY (steps: PICK, GOODS_ISSUE):

```

DRAFT → RELEASED → PICKING → PICKED → GOODS_ISSUED → COMPLETED
                                   ↑ skips PACKED, STAGED, APPROVAL
                                   ↑ skips SHIP statuses

```

PICK_AND_SHIP (steps: PICK, GOODS_ISSUE, SHIP):

```

DRAFT → RELEASED → PICKING → PICKED → GOODS_ISSUED → IN_TRANSIT → ... → DELIVERED
→ COMPLETED
                                   ↑ skips PACKED, STAGED, APPROVAL

```

PICK_PACK_SHIP (steps: PICK, PACK, GOODS_ISSUE, SHIP):

```

DRAFT → RELEASED → PICKING → PICKED → PACKED → GOODS_ISSUED → IN_TRANSIT → ... →
DELIVERED → COMPLETED
                                   ↑ skips STAGED, APPROVAL

```

Full route with all steps (custom):

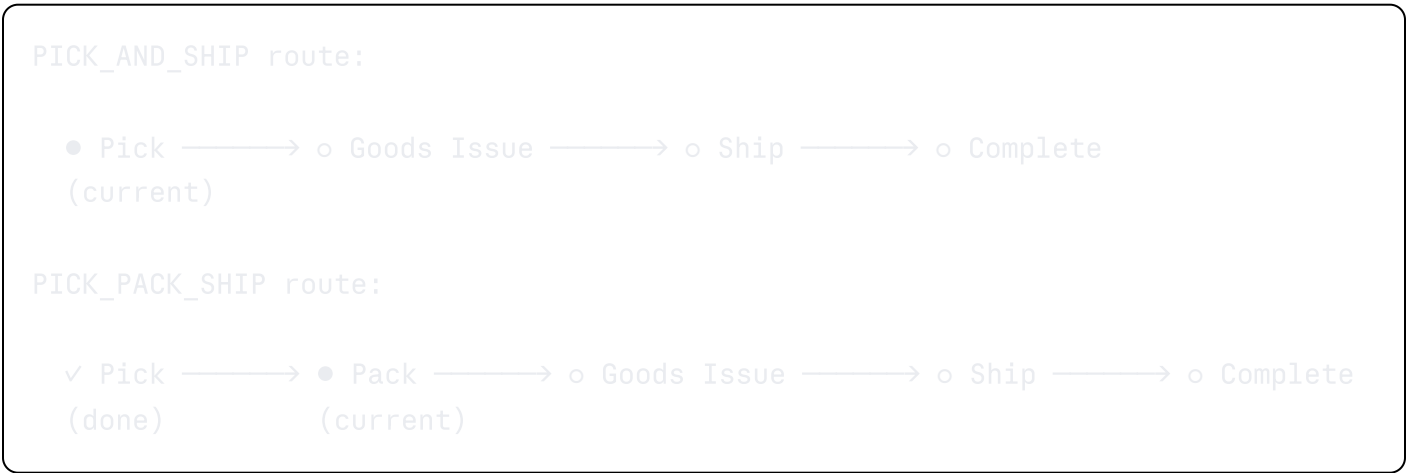
```

DRAFT → RELEASED → PICKING → PICKED → PACKED → STAGED → PENDING_APPROVAL →
GOODS_ISSUED → IN_TRANSIT → ... → DELIVERED → COMPLETED

```

7.5 UI — Delivery Order Step Tracker

The delivery order detail page shows a visual step tracker that only includes the steps in the delivery's route. Skipped steps do not appear:



7.6 Cancellation

Cancellation (→ CANCELLED) is available from any status regardless of route, following existing behavior. If the delivery has a linked Sale Order, the SO's delivery status is recalculated.

8. Database Migrations

Four migrations are required for this addendum. They are additive and non-breaking.

8.1 Migration Order

#	Migration Name	Schema	Type	Description
M1	CreateFulfillmentRoutes	logistics	CREATE TABLE	FulfillmentRoutes + FulfillmentRouteSteps + seed data
M2	AddFulfillmentRouteToProductVariants	inventory	ALTER TABLE	Add fulfillment_route_id FK to ProductVariants
M3	AddFulfillmentRouteToSaleOrderLines	demand	ALTER TABLE	Add fulfillment_route_id FK to SaleOrderLines
M4	ExtendDeliveryOrdersForSaleOrder	logistics	ALTER TABLE	Add sale_order_id, fulfillment_route_id to delivery_orders; add sale_order_line_id to delivery_order_lines

8.2 Migration Details

M1 — CreateFulfillmentRoutes

sql


```
-- See §3.5 for seed route definitions
-- The first org's seed routes serve as the template:
--
-- Route 1: PICK_ONLY (is_default=0, is_system=1, requires_packing=0, requires_shipping=0)
--   Steps: (1, PICK), (2, GOODS_ISSUE)
--
-- Route 2: PICK_AND_SHIP (is_default=1, is_system=1, requires_packing=0, requires_shipping=0)
--   Steps: (1, PICK), (2, GOODS_ISSUE), (3, SHIP)
--
-- Route 3: PICK_PACK_SHIP (is_default=0, is_system=1, requires_packing=1, requires_shipping=0)
--   Steps: (1, PICK), (2, PACK), (3, GOODS_ISSUE), (4, SHIP)
```

M2 — AddFulfillmentRouteToProductVariants

sql

```
ALTER TABLE inventory.ProductVariants
ADD fulfillment_route_id INT NULL
    CONSTRAINT FK_ProductVariants_FulfillmentRoute
    FOREIGN KEY REFERENCES logistics.FulfillmentRoutes(fulfillment_route_id);

CREATE INDEX IX_ProductVariants_FulfillmentRoute
    ON inventory.ProductVariants(fulfillment_route_id)
    WHERE fulfillment_route_id IS NOT NULL;

-- No backfill needed – existing variants start with NULL (no default route)
-- Organizations should assign routes via bulk assignment or one-by-one
```

M3 — AddFulfillmentRouteToSaleOrderLines

sql

```
ALTER TABLE demand.SaleOrderLines
ADD fulfillment_route_id INT NULL
    CONSTRAINT FK_SaleOrderLines_FulfillmentRoute
    FOREIGN KEY REFERENCES logistics.FulfillmentRoutes(fulfillment_route_id);

CREATE INDEX IX_SaleOrderLines_FulfillmentRoute
    ON demand.SaleOrderLines(fulfillment_route_id)
    WHERE fulfillment_route_id IS NOT NULL;

-- No backfill needed – existing SO lines pre-date route requirement
-- The confirmation gate (§5.4) only applies to NEW confirmations going forward
```

M4 — ExtendDeliveryOrdersForSaleOrder

sql

```

-- delivery_orders: add sale_order_id and fulfillment_route_id
ALTER TABLE logistics.delivery_orders
ADD sale_order_id BIGINT NULL
    CONSTRAINT FK_DeliveryOrders_SaleOrder
    FOREIGN KEY REFERENCES demand.SaleOrders(sale_order_id);

ALTER TABLE logistics.delivery_orders
ADD fulfillment_route_id INT NULL
    CONSTRAINT FK_DeliveryOrders_FulfillmentRoute
    FOREIGN KEY REFERENCES logistics.FulfillmentRoutes(fulfillment_route_id);

CREATE INDEX IX_DeliveryOrders_SaleOrder
    ON logistics.delivery_orders(sale_order_id)
    WHERE sale_order_id IS NOT NULL;

CREATE INDEX IX_DeliveryOrders_Route
    ON logistics.delivery_orders(fulfillment_route_id)
    WHERE fulfillment_route_id IS NOT NULL;

-- delivery_order_lines: add sale_order_line_id
ALTER TABLE logistics.delivery_order_lines
ADD sale_order_line_id BIGINT NULL
    CONSTRAINT FK_DOLines_SaleOrderLine
    FOREIGN KEY REFERENCES demand.SaleOrderLines(sale_order_line_id);

CREATE INDEX IX_DOLines_SaleOrderLine
    ON logistics.delivery_order_lines(sale_order_line_id)
    WHERE sale_order_line_id IS NOT NULL;

```

8.3 Rollback

Migration Rollback Steps

M1	DROP TABLE logistics.FulfillmentRouteSteps, logistics.FulfillmentRoutes (after clearing FKs from M2, M3, M4)
M2	DROP COLUMN fulfillment_route_id from ProductVariants (after clearing FK and index)
M3	DROP COLUMN fulfillment_route_id from SaleOrderLines (after clearing FK and index)
M4	DROP COLUMN sale_order_line_id from delivery_order_lines; DROP COLUMN sale_order_id, fulfillment_route_id from delivery_orders (after clearing FKs and indexes)

9. API Changes

9.1 New Endpoints — Fulfillment Routes

Endpoint	Method	Description	Permission
/api/fulfillment-routes	GET	List active routes for org (with steps)	authenticated
/api/fulfillment-routes	POST	Create a new route with steps	admin
/api/fulfillment-routes/{id}	GET	Get route with steps	authenticated
/api/fulfillment-routes/{id}	PUT	Update route name, description, steps	admin
/api/fulfillment-routes/{id}/deactivate	PATCH	Soft-delete a route (must not be in use)	admin
/api/fulfillment-routes/{id}/set-default	PATCH	Set as org default route	admin
/api/fulfillment-routes/{id}/assign-by-category	POST	Bulk assign to variants in a product category	admin

9.2 Modified Endpoints — Product Variants

Endpoint	Change	Change ID
GET /api/product-variants/{id}	Return fulfillment_route_id and route summary in response	C2
PUT /api/product-variants/{id}	Accept fulfillment_route_id in update DTO	C2
GET /api/product-variants (list)	Include fulfillment_route_code and fulfillment_route_name in list DTO	C2

9.3 Modified Endpoints — Sale Orders

Endpoint	Change	Change ID
POST /api/sale-orders/{id}/lines	Accept optional fulfillment_route_id per line	C3
PUT /api/sale-orders/{id}/lines/{lineId}	Accept optional fulfillment_route_id update	C3
GET /api/sale-orders/{id}/lines	Return effective_route_id, effective_route_code, route_source (VARIANT/LINE_OVERRIDE/ORG_DEFAULT/NONE) per line	C3
POST /api/sale-orders/{id}/confirm	Validate all lines have routes; on success create delivery orders grouped by route	C3, C4
GET /api/sale-orders/{id}/delivery-preview	Return delivery grouping preview (route groups, line counts) without creating anything	C3

9.4 Modified Endpoints — Delivery Orders

Endpoint	Change	Change ID
GET /api/delivery-orders/{id}	Return fulfillment_route_id, fulfillment_route_code, sale_order_id, allowed_transitions (route-aware)	C4, C5
POST /api/delivery-orders/{id}/advance	Use route-aware state machine to determine next status	C5
GET /api/delivery-orders (list)	Support filter by from_source_type=SALE_ORDER, sale_order_id	C4

10. UI Wireframes & Specifications

10.1 Fulfillment Route Management

Settings → Fulfillment Routes

[+ New Route]

Code	Name	Steps	Default	
PICK_ONLY	Pick Only	Pick→GI	☐	
PICK_AND_SHIP	Pick & Ship	Pick→GI→Ship	☒	
PICK_PACK_SHIP	Pick Pack Ship	Pick→Pk→GI→S	☐	
HIGH_VALUE	High Value Full	Pk→Pk→St→Ap →GI→Ship	☐	

= System route (cannot be deleted)

= Organization default route

10.2 Fulfillment Route Editor

Fulfillment Route: PICK_PACK_SHIP

Code	Name
PICK_PACK_SH	Pick, Pack & Ship

Description

Full cycle – items picked, packed into boxes/pallets, then shipped via carrier to customer

Steps (drag to reorder):

① PICK	☒ Mandatory	Generate pick list
② PACK	☒ Mandatory	Pack into HUs
③ GOODS_ISSUE	☒ Mandatory	Post goods issue
④ SHIP	☒ Mandatory	Book carrier, ship
[+ Add Step]		
Available: STAGE, APPROVAL		

Preview: DRAFT → RELEASED → PICKING → PICKED → PACKED
→ GOODS_ISSUED → IN_TRANSIT → ... → DELIVERED
→ COMPLETED

☐ Set as organization default

[Save] [Cancel]

10.3 Delivery Order — Route-Aware Step Tracker

Delivery: DO-20261003-002

Sale Order: SO-1085

Route: Pick Pack Ship

Customer: Punjab Group

✓ Pick → ● Pack → ○ Goods Issue → ○ Ship

(done) (current)

Lines

	Product	Qty	Picked	Packed	Shipped
1	Copper Wire	600	600	0	0

[Start Packing]

10.4 Sale Order Fulfillment Tab

Sale Order: SO-1085

Status: CONFIRMED

[Header] [Lines] [Fulfillment] [Attachments] [History]

▲ active

Delivery Orders (3)

Delivery	Route	Lines	Status	Action
D0-20261003-001	Pick Only	1	● Picked	[View]
└ Steel Pipes × 500 – customer collecting today				
D0-20261003-002	Pick Pack Ship 1		● Packing	[View]
└ Copper Wire × 600 – packing in progress				
D0-20261003-003	Pick & Ship	1	● Picking	[View]
└ Synth Hyd Oil × 100 – pick list generated				

Overall Fulfillment: 0 of 3 lines delivered

0%

11. Business Rules

Consolidated business rules for all 5 changes.

Rule ID	Change	Rule Description	Enforcement Point
BR-C1-01	C1	Route code must be unique per org	FulfillmentRouteService
BR-C1-02	C1	At most one route per org can have is_default = 1	FulfillmentRouteService
BR-C1-03	C1	Every route must have at least PICK and GOODS_ISSUE steps	FulfillmentRouteService
BR-C1-04	C1	Step order must be sequential starting from 1, no gaps	FulfillmentRouteService
BR-C1-05	C1	PICK must always be the first step; GOODS_ISSUE must come before SHIP	FulfillmentRouteService
BR-C1-06	C1	System routes (is_system = 1) cannot be deleted, only deactivated	FulfillmentRouteService
BR-C1-07	C1	A route cannot be deactivated while assigned to any active product variant or open SO line	FulfillmentRouteService
BR-C1-08	C1	Seed routes are inserted per org on tenant provisioning	TenantProvisioningService
BR-C1-09	C1	requires_packing and requires_shipping are computed from steps on save (PACK step → requires_packing = 1, SHIP step → requires_shipping = 1)	FulfillmentRouteService

Rule ID	Change	Rule Description	Enforcement Point
BR-C2-01	C2	Product variant's fulfillment_route_id must reference an active route in the same org	ProductVariantService
BR-C2-02	C2	Bulk assignment by category only sets NULL variants, does not overwrite existing assignments	FulfillmentRouteService
BR-C3-01	C3	SO line fulfillment_route_id (when set) must reference an active route in the same org	SaleOrderLineService
BR-C3-02	C3	SO confirmation (DRAFT → CONFIRMED) requires every line to resolve an effective route	SaleOrderService
BR-C3-03	C3	Effective route resolution order: SO line override → variant default → org default → NULL (blocks)	EffectiveRouteResolver
BR-C3-04	C3	SO line route can be changed while SO is in DRAFT; locked after confirmation	SaleOrderLineService
BR-C3-05	C3	Delivery preview is computed on the fly and not persisted — changes as lines are edited	SaleOrderDeliveryPreviewService
BR-C4-01	C4	SO confirmation creates delivery orders automatically, grouped by effective route	SaleOrderDeliveryCreator
BR-C4-02	C4	Lines with the same effective route go into the same delivery order	SaleOrderDeliveryCreator
BR-C4-03	C4	Delivery orders are created in DRAFT status — require explicit release for picking	SaleOrderDeliveryCreator

Rule ID	Change	Rule Description	Enforcement Point
BR-C4-04	C4	delivery_orders.from_source_type = 'SALE_ORDER' for all SO-originated deliveries	SaleOrderDeliveryCreator
BR-C4-05	C4	Each delivery_order_line links back to its sale_order_line_id for traceability	SaleOrderDeliveryCreator
BR-C4-06	C4	Cancelling a SO cancels all linked delivery orders that are not yet GOODS_ISSUED	SaleOrderService
BR-C4-07	C4	A delivery order past GOODS_ISSUED cannot be auto-cancelled by SO cancellation — requires manual reversal	SaleOrderService
BR-C5-01	C5	Delivery state machine skips statuses whose step_code is not in the delivery's fulfillment route	RouteAwareDeliveryStateMachine
BR-C5-02	C5	PICK and GOODS_ISSUE steps are always required — routes without them are rejected at creation	FulfillmentRouteService
BR-C5-03	C5	The "Advance" action on a delivery determines the next valid status based on the route, not the full 15-status chain	DeliveryOrderService
BR-C5-04	C5	Delivery order detail UI shows only the steps in the delivery's route (step tracker)	Delivery Detail UI
BR-C5-05	C5	Cancellation (→ CANCELLED) is always available regardless of route	DeliveryOrderService

12. Test Scenarios

Test ID	Change	Scenario	Expected Result
T-C1-01	C1	Create fulfillment route with PICK and GOODS_ISSUE steps	Route created, requires_packing = 0, requires_shipping = 0
T-C1-02	C1	Create route with PICK, PACK, GOODS_ISSUE, SHIP	Route created, requires_packing = 1, requires_shipping = 1
T-C1-03	C1	Create route without PICK step	Validation error: PICK step is required
T-C1-04	C1	Create route without GOODS_ISSUE step	Validation error: GOODS_ISSUE step is required
T-C1-05	C1	Create route with SHIP before GOODS_ISSUE	Validation error: GOODS_ISSUE must precede SHIP
T-C1-06	C1	Create route with duplicate code in same org	Validation error: code must be unique per org
T-C1-07	C1	Set route as org default when another default exists	Previous default cleared, new one set
T-C1-08	C1	Deactivate route assigned to active variant	Validation error: route is in use
T-C1-09	C1	Delete system route	Validation error: system routes cannot be deleted
T-C2-01	C2	Assign fulfillment route to product variant	fulfillment_route_id saved on variant
T-C2-02	C2	Assign route from different org	Validation error: route must belong to same org
T-C2-03	C2	Bulk assign route by category (3 variants, 1 already has route)	2 variants updated, 1 skipped (already assigned)
T-C2-04	C2	Clear route from variant (set to NULL)	fulfillment_route_id = NULL

Test ID	Change	Scenario	Expected Result
T-C3-01	C3	Add SO line for variant with route — no line override	Effective route = variant's route, source = VARIANT
T-C3-02	C3	Override route on SO line	Effective route = line override, source = LINE_OVERRIDE
T-C3-03	C3	SO line variant has no route, org has default	Effective route = org default, source = ORG_DEFAULT
T-C3-04	C3	SO line variant has no route, no org default, no line override	Effective route = NULL, source = NONE
T-C3-05	C3	Confirm SO with all lines having effective routes	SO confirmed, delivery orders created
T-C3-06	C3	Confirm SO with 1 line missing route	Validation error: line {N} has no fulfillment route
T-C3-07	C3	Confirm SO with line pointing to inactive route	Validation error: route is inactive
T-C3-08	C3	Change line route after SO confirmed	Validation error: SO is confirmed, route locked
T-C3-09	C3	Delivery preview shows correct grouping	Preview returns 3 groups for 3 different routes
T-C4-01	C4	Confirm SO with 3 lines, all same route	1 delivery order created with 3 lines
T-C4-02	C4	Confirm SO with 3 lines, 3 different routes	3 delivery orders created, 1 line each
T-C4-03	C4	Confirm SO with 3 lines, 2 same route + 1 different	2 delivery orders (2 lines + 1 line)
T-C4-04	C4	Delivery order has from_source_type = SALE_ORDER	Verified on created delivery

Test ID	Change	Scenario	Expected Result
T-C4-05	C4	Delivery order lines link back to SO lines	sale_order_line_id populated on each DO line
T-C4-06	C4	Cancel SO — deliveries in DRAFT/RELEASED	All linked deliveries cancelled
T-C4-07	C4	Cancel SO — one delivery already GOODS_ISSUED	DRAFT/RELEASED deliveries cancelled; GOODS_ISSUED delivery remains (manual reversal needed)
T-C5-01	C5	Advance PICK_ONLY delivery from PICKED	Next status = GOODS_ISSUED (skips PACKED, STAGED, APPROVAL)
T-C5-02	C5	Advance PICK_ONLY delivery from GOODS_ISSUED	Next status = COMPLETED (skips all SHIP statuses)
T-C5-03	C5	Advance PICK_AND_SHIP delivery from PICKED	Next status = GOODS_ISSUED (skips PACKED, STAGED, APPROVAL)
T-C5-04	C5	Advance PICK_AND_SHIP delivery from GOODS_ISSUED	Next status = IN_TRANSIT
T-C5-05	C5	Advance PICK_PACK_SHIP delivery from PICKED	Next status = PACKED
T-C5-06	C5	Advance PICK_PACK_SHIP delivery from PACKED	Next status = GOODS_ISSUED (skips STAGED, APPROVAL)
T-C5-07	C5	Full route (all steps) — advance from PACKED	Next status = STAGED
T-C5-08	C5	Cancel delivery from any status (any route)	Status = CANCELLED
T-C5-09	C5	Step tracker UI for PICK_ONLY	Shows: Pick → Goods Issue → Complete (3 steps only)
T-C5-10	C5	Step tracker UI for PICK_PACK_SHIP	Shows: Pick → Pack → Goods Issue → Ship → Complete (4 steps)

13. Development Phases

13.1 Phase Breakdown

Phase	Changes	Estimated Days	Dependencies
Phase A — Route Configuration	C1 (FulfillmentRoutes entity, steps, API, seed data, UI)	3 days	None
Phase B — Variant Assignment	C2 (ProductVariants FK, variant form UI, bulk assign)	2 days	Phase A
Phase C — SO Route Selection & Gate	C3 (SaleOrderLines FK, route resolver, confirmation validator, SO line UI, delivery preview)	4 days	Phase A, B
Phase D — Delivery Creation from SO	C4 (delivery grouping logic, delivery_orders extension, SO-to-DO creation, fulfillment tab UI)	4 days	Phase C
Phase E — Route-Aware State Machine	C5 (state machine modifications, step tracker UI, advance logic)	3 days	Phase D
Phase F — Testing & Integration	All changes — integration tests, E2E scenarios	3 days	Phase A-E
TOTAL		19 days	

13.2 Parallelization Opportunities

- Phase B (Variant Assignment) can start immediately after Phase A's migration is done — the UI work can run in parallel with Phase C's backend
- Phase C backend (route resolver, validator) and Phase C frontend (SO line UI) can run in parallel
- Phase E (state machine) is independent of Phase D's UI — backend can start once D's migration is applied

13.3 New Permission Claims Summary

Claim Code	Module	Description
fulfillment_route_view	Logistics	View fulfillment routes
fulfillment_route_manage	Logistics	Create, edit, deactivate fulfillment routes
fulfillment_route_assign	Logistics	Assign routes to product variants (including bulk)

Appendix A — Entity Relationship Summary



Appendix B — Deferred to Future Addendums

Topic	Description
Partial Delivery & Backorders	When only part of a DO line is picked/shipped — auto-create backorder delivery
Delivery Notes & Print	Delivery note PDF generation, packing slip print
Carrier Integration for SO Deliveries	Extending ICourierProvider for SO-originated consignments (existing infra, needs wiring)
Invoice Generation from Delivery	Creating invoices when delivery reaches GOODS_ISSUED or DELIVERED
Returns & Reverse Logistics	Customer returns, credit notes, return delivery orders
Warehouse Zone Routing	Route steps that specify which warehouse zone/area the product moves through
Wave Picking	Grouping multiple delivery pick lists into a single warehouse wave

End of FSD Addendum 33 — Fulfillment Routes & Sale Order Delivery Integration